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Arbitrage-Free Implied Volatility Surface Construction: From Parametric Models to Neural Operators

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Declaration

The work contained in this thesis is my own work unless otherwise stated.

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I would like to thank my supervisor for their guidance throughout this project.

Abstract

The implied volatility surface is the central object of equity derivatives markets, yet it is only ever observed as a sparse, noisy and irregular cloud of option quotes. Reconstructing a complete, smooth and arbitrage-free surface from these observations is an ill-posed inverse problem, and the quality of the reconstruction directly affects pricing, hedging and risk management. This thesis studies the problem across the full methodological spectrum, on a single unified S&P 500 dataset built from OptionMetrics. We first replicate the parametric benchmark of Gatheral and Jacquier—the SVI and SSVI families, their no-arbitrage theory and calibration recipes—validating every component against the original paper. We then implement a derivative-constrained deep smoother in the spirit of Ackerer et al. and Hoshisashi et al., in which a neural corrector multiplies an SSVI prior and no-arbitrage conditions are enforced through automatic differentiation, and finally a neural operator following the operator deep smoothing approach, which learns the smoothing map itself and applies to unseen days without retraining. All methods are evaluated under a common protocol: hold-out reconstruction error, a systematic static-arbitrage audit based on the Durrleman condition, robustness to quote sparsity, and behaviour across market regimes including the COVID-19 crash. As a contribution, we quantify the low-data regime of operator smoothing available to academic research and measure whether pre-training on synthetic SSVI surfaces closes the gap to the data-rich industrial setting.

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Introduction

When an equity option is priced with the Black–Scholes formula, a single parameter is not directly observable: the volatility σ . The *implied volatility* of a quoted option is the value of σ that reproduces its market price, and when computed across strikes K and expiries τ it traces out the *implied volatility surface* $(k, \tau) \mapsto \sigma_{\text{IV}}(k, \tau)$, with $k = \log(K/F)$ the log-moneyness against the forward. The surface is the market’s summary of the risk-neutral distribution of the underlying at every horizon; it is the input to exotic pricing, hedging and risk systems, and its shape—the skew, the term structure—is economically informative in its own right.

The market, however, never exhibits a surface. It exhibits a *sparse, noisy, irregular* cloud of quotes: only certain strikes and expiries trade, liquidity concentrates near the money, and bid–ask spreads widen rapidly in the wings. Reconstructing a complete, smooth surface from these observations is an ill-posed inverse problem: infinitely many surfaces interpolate the same quotes. The discipline that selects an admissible one is *absence of static arbitrage*: the reconstructed prices must not allow riskless profit through butterfly or calendar strategies. These constraints admit a precise analytical form—non-negativity of the Durrleman function g for butterfly arbitrage, monotonicity of total variance in maturity for calendar arbitrage—which makes them both testable and, crucially for this thesis, *trainable*.

Three generations of methods address the problem. Parametric families, culminating in SVI and SSVI [5], offer interpretability and provable no-arbitrage guarantees at the cost of rigidity. Per-surface neural smoothers [1, 2, 6] recover flexibility by penalising arbitrage through the network’s automatically-differentiated derivatives, but require one model per surface. Neural operators [11] learn the smoothing *map* itself: trained once across many days, a single discretisation-invariant model smooths any quote set without retraining.

Contributions. This thesis makes four contributions.

1. **A unified benchmark.** All three generations are implemented and evaluated on the *same* daily S&P 500 dataset (OptionMetrics, 2018–2022), under the *same* protocol: hold-out reconstruction error, a dense-grid static-arbitrage audit, robustness to quote sparsity, and per-regime analysis. Published papers each use their own data and metrics; to our knowledge no like-for-like comparison exists.
2. **A validated replication.** The complete Gatheral–Jacquier toolkit—three SVI parameterisations and their exact conversions, the butterfly-elimination algorithm, the calibration recipes, the SSVI no-arbitrage theorems, arbitrage-free interpolation—is replicated and validated against the paper’s own numerical examples, correcting in passing a typographical omission in the printed statement of the Jump–Wings inversion (Lemma 3.2).
3. **A systematic arbitrage audit.** Beyond fit error, every method is scored on *how much* static arbitrage it leaves on real data—violation rates and magnitudes of the Durrleman condition on dense grids, cross-slice crossedness, and term-structure inversions—quantities the literature rarely reports.
4. **The low-data operator experiment.** Operator deep smoothing was demonstrated on ~ 60 million intraday quotes; academic data access is daily, two orders of magnitude smaller.

We measure operator performance in this regime and test whether pre-training on synthetic SSVI surfaces closes the gap (regimes R1/R2/R3: real-only, synthetic-only, pre-train-then-fine-tune).

Outline. Chapter 1 sets the financial and mathematical foundations. Chapter 2 reviews the three methodological generations. Chapter 3 documents the data pipeline. Chapters 4–6 develop the three method families, each validated on synthetic ground truth before being applied to market data. Chapter 7 performs the unified comparison, and the conclusion discusses limitations and extensions.

Chapter 1

Foundations

1.1 Options, Black–Scholes and implied volatility

Let S denote the underlying, $F = F(t, T)$ its forward for delivery at T , and $D = D(t, T)$ the discount factor. Under the Black (forward) model, the price of a European call of strike K and time-to-maturity $\tau = T - t$ is

$$C(K, \tau; \sigma) = D [F N(d_+) - K N(d_-)], \quad d_{\pm} = \frac{\log(F/K)}{\sigma\sqrt{\tau}} \pm \frac{\sigma\sqrt{\tau}}{2}, \quad (1.1)$$

with N the standard normal cdf. Since $\partial C / \partial \sigma > 0$ (the vega), Equation (1.1) can be inverted: the *implied volatility* $\sigma_{\text{IV}}(K, \tau)$ of a quote C^{mkt} is the unique root of $C(K, \tau; \sigma) = C^{\text{mkt}}$. The empirical fact that σ_{IV} varies with (K, τ) —the *smile* and the term structure—contradicts the constant- σ assumption and gives rise to the implied volatility surface.

1.2 Coordinates: log-moneyness and total variance

Two changes of variable make the no-arbitrage theory tractable. The *log-moneyness* $k = \log(K/F)$ centres the smile on the forward. The *total variance*

$$w(k, \tau) = \sigma_{\text{IV}}^2(k, \tau) \tau \quad (1.2)$$

is the natural vertical coordinate: both static no-arbitrage conditions take their simplest form in w .

1.3 Static no-arbitrage

Following [5, 10], a surface is free of static arbitrage if and only if each maturity slice is free of butterfly arbitrage and the family of slices is free of calendar arbitrage.

Definition 1.3.1 (Butterfly arbitrage and the Durrleman function). For a slice $w(\cdot)$ at fixed maturity, define

$$g(k) = \left(1 - \frac{k w'(k)}{2 w(k)}\right)^2 - \frac{w'(k)^2}{4} \left(\frac{1}{w(k)} + \frac{1}{4}\right) + \frac{w''(k)}{2}. \quad (1.3)$$

The slice is free of butterfly arbitrage if and only if $g(k) \geq 0$ for all k and call prices tend to zero at extreme strikes (for SVI, the wing-slope condition $b(1 + |\rho|) < 2$).

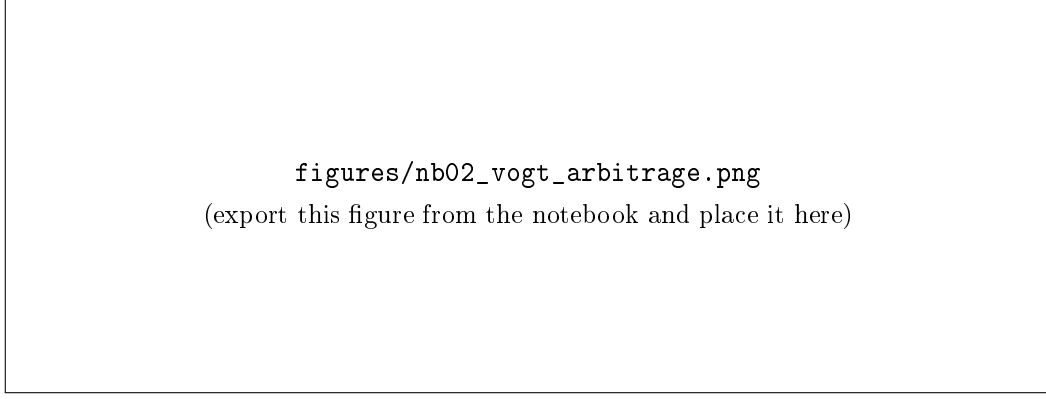


Figure 1.1: The Vogt SVI parameters of [5, Ex. 3.1]. The total-variance slice (left) looks innocent, yet the Durrleman function dips below zero (centre) and the implied risk-neutral density turns negative (right): a genuine butterfly arbitrage invisible to the naked eye. Figure produced by our replication (Notebook 02).

The financial content of Equation (1.3) is the Breeden–Litzenberger identity: the risk-neutral density is proportional to $\partial^2 C / \partial K^2$, and in total-variance coordinates

$$p(k) = \frac{g(k)}{\sqrt{2\pi w(k)}} \exp\left(-\frac{1}{2}d_-(k)^2\right), \quad d_-(k) = -\frac{k}{\sqrt{w(k)}} - \frac{\sqrt{w(k)}}{2}, \quad (1.4)$$

so $g < 0$ is exactly a negative implied density.

Definition 1.3.2 (Calendar arbitrage). The surface is free of calendar arbitrage if and only if $\tau \mapsto w(k, \tau)$ is non-decreasing for every k ; equivalently, total-variance slices never cross.

Equation (1.3) involves only w , w' and w'' . This is the property exploited throughout the thesis: for parametric models the derivatives are closed-form; for neural models they are obtained *exactly* by automatic differentiation, turning the no-arbitrage conditions into differentiable training penalties.

1.4 The reconstruction problem

Given a day's quotes $\{(k_i, \tau_i, \sigma_{IV_i})\}_{i=1}^p$, the problem studied in this thesis is to construct $\hat{w} : \mathbb{R} \times (0, \infty) \rightarrow (0, \infty)$ that (i) fits the quotes, (ii) satisfies the conditions of Section 1.3, and (iii) generalises to quotes not used in the construction. Conditions (i)–(iii) conflict; quantifying the trade-off across method families is the object of the thesis.

Chapter 2

Related Work

2.1 Parametric surfaces: SVI and SSVI

The Stochastic Volatility Inspired (SVI) parameterisation [4] models a total-variance slice with five parameters,

$$w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right), \quad (2.1)$$

and owes its longevity to closed-form derivatives, trader-interpretable reparameterisations, and its exact large-maturity consistency with the Heston smile. Gatheral and Jacquier [5] supply the modern no-arbitrage theory: explicit butterfly conditions via the Durrleman function, three equivalent parameterisations (raw, natural, Jump–Wings) with exact conversions, an algorithm that *eliminates* butterfly arbitrage by adjusting the Jump–Wings call slope and minimum variance, and the surface-level SSVI family

$$w(k, \theta_\tau) = \frac{\theta_\tau}{2} \left(1 + \rho\varphi(\theta_\tau)k + \sqrt{(\varphi(\theta_\tau)k + \rho)^2 + 1 - \rho^2} \right), \quad (2.2)$$

where θ_τ is the at-the-money total variance and φ a decreasing function, with sharp theorems characterising when Equation (2.2) is entirely free of static arbitrage. Calibration is delicate—the SVI objective has spurious local minima—and the quasi-explicit reformulation of [3] restores convexity of the inner subproblem. Chapter 4 replicates this toolkit in full.

2.2 Per-surface neural smoothers

Ackerer, Tagasovska and Vatter [1] introduced the architecture adopted in Chapter 5: the implied-volatility (or total-variance) surface is modelled as the *product of a prior and a neural corrector*, and the no-arbitrage conditions are imposed as *soft constraints*—penalties on the network’s derivatives evaluated on a grid wider than the quotes. Chataigner, Crépey and Dixon [2] work in price space, penalising the Dupire-relevant derivatives ($\partial_{KK}^2 P \geq 0$, $\partial_T P \geq 0$) and extracting local volatility by automatic differentiation; importantly they document that *hard*-constrained architectures (e.g. splines of non-negative weights) lose too much expressiveness, motivating the soft-constraint consensus. Hoshisashi, Phelan and Barucca [6] systematise the approach (DCNN): exact first- and second-order derivatives via automatic differentiation, hinge penalties for each no-arbitrage inequality evaluated on a collocation set distinct from the data, and twice-differentiable activations—ReLU being excluded since its second derivative vanishes almost everywhere.

2.3 Neural operators

Neural operators [7] learn maps between function spaces with the defining property of *discretisation invariance*: the same trained model accepts inputs sampled on arbitrary grids. Operator

deep smoothing [11] applies a graph neural operator (GNO) to implied-volatility smoothing: each day’s quotes are treated as a pointwise discretisation of a latent surface, and a single $\sim 100\text{k}$ -parameter model, trained on roughly a decade of 20-minute CBOE S&P 500 snapshots (~ 60 million quotes), smooths any quote set in one forward pass. The architecture modifies the standard GNO by making the first layer purely non-local, which allows the hidden state to be *created* at arbitrary output locations (interpolation); training uses a vega-weighted relative fitting loss and finite-difference no-arbitrage penalties. Chapter 6 adapts this approach to the daily-data regime available to academic research.

2.4 Positioning

The three families have never been compared on equal terms: [5] predates the neural literature, [1] and [6] benchmark against their own SVI implementations on different datasets, and [11] compares to Akerer’s *reported* numbers rather than a shared sample. This thesis contributes the missing controlled comparison, together with an arbitrage audit of uniform stringency and the low-data transfer study of Chapter 6.

Chapter 3

Data

3.1 Sources

All market data are from OptionMetrics IvyDB US, accessed through WRDS. Four tables are used: daily option quotes on the S&P 500 index (SPX), forward prices, the zero-coupon yield curve, and—as an industry reference only—OptionMetrics’ own smoothed volatility surface. The sample covers 2018–2022: a calm regime (2019), the COVID-19 crash (March 2020), the recovery, and the 2022 sell-off, enabling the per-regime analysis of Chapter 7.

3.2 Cleaning pipeline

The raw exports are processed with a lazy, streaming Polars pipeline (Notebook 01) whose main steps are:

1. **Typing and units.** OptionMetrics stores strikes in thousandths (a displayed strike of 4500 is stored as 4500000); dates are parsed defensively; quotes with $\text{bid} \leq 0$, $\text{ask} < \text{bid}$, or maturities outside $[7, 730]$ days are discarded.
2. **Missing implied volatilities.** The fields `impl_volatility`, `delta` and `vega` are missing on exactly the same rows ($\sim 5.7\%$); 99.98% of these are deep in-the-money options for which OptionMetrics does not attempt the inversion (the time value is buried in the spread). Since the surface is built from out-of-the-money quotes, these rows are dropped; the 118 OTM exceptions (out of 8.5 million rows) are immaterial.
3. **Contract types.** Standard SPX (third-Friday, AM-settled) and SPXW weeklys (PM-settled) may share an expiration date with slightly different prices. The `root` field being unpopulated in the export, contracts are labelled by the settlement flag, and third-Friday collisions are resolved by preferring the standard contract; weeklys are retained on all other expirations, densifying the short end. One expiration therefore maps to exactly one contract.
4. **Coordinates and discounting.** Forwards are taken from the quote file and completed by a join with the forward table; $k = \log(K/F)$; τ in ACT/365; the discount factor $D = e^{-r\tau}$ uses the zero curve at the nearest tenor.
5. **Validation.** Implied volatilities are recomputed independently by a vectorised Black-inversion for every retained quote and compared to OptionMetrics’ values: the median absolute gap is below 10^{-3} in volatility, validating forwards, discounting and coordinates simultaneously.

`figures/nb01_quote_density.png`
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Figure 3.1: Number of out-of-the-money SPX quotes per trading day after cleaning. The weekly sawtooth reflects expiration cycles.

The output is a typed Parquet dataset (one row per quote: date, expiry, τ , k , bid/ask/mid, spread, volume, open interest, implied volatility, discount) on which all subsequent chapters operate; slices are always formed by exact expiration date, never by rounded maturity.

3.3 Evaluation domains

Two evaluation domains are carried through the thesis. The *full* domain retains every cleaned quote and is honest about real difficulty; the *paper* domain restricts to $k \in [-0.6, 0.4]$ and $\text{DTE} \geq 10$, matching the effective domains of the reference papers and enabling literature-comparable numbers. Results are reported on both.

Chapter 4

The Parametric Benchmark: SVI and SSVI

Before any neural method can claim an improvement, the parametric state of the art must be implemented *properly* and audited *honestly*. This chapter replicates the complete toolkit of [5] and turns it into a quantified benchmark on the real SPX dataset of Chapter 3. Every replicated component is validated against the numbers printed in the original paper.

4.1 Replication of Gatheral–Jacquier (2014)

4.1.1 Parameterizations and exact conversions

We implement the three equivalent SVI parameterizations: *raw* (Equation (2.1)), *natural*, and *jump-wings* (JW), together with the closed-form conversions of Lemmas 3.1–3.2 of the paper. The JW parameters $(v_t, \psi_t, p_t, c_t, \tilde{v}_t)$ — ATM variance, ATM skew, wing slopes and minimum variance — are the trader-facing coordinates and the vehicle of the butterfly-elimination algorithm below.

Remark 4.1.1 (A correction to the printed Lemma 3.2). When inverting the JW map back to raw parameters we found that the printed formula omits one term; using it verbatim fails to recover the raw parameters of the paper’s own example. Appendix A.1 derives the corrected inverse, which handles the case $m < 0$ through the identity $\sqrt{m^2 + \sigma^2} = m/\beta$. All conversions were validated by round-trips over 2,000 random parameter draws, with maximum error 2.7×10^{-10} , and the JW values of the paper’s Vogt example (0.01742625, -0.1752111 , 0.6997381, 1.316798, 0.0116249) are reproduced to the printed digit.

4.1.2 Static-arbitrage diagnostics and the Vogt smile

The butterfly criterion is the Durrleman condition $g(k) \geq 0$ of Equation (1.3) together with the wing condition $b(1 + |\rho|) < 2$ (Lee’s moment bound [8]); calendar arbitrage between two slices is measured by their *crossedness*. Applied to the Vogt smile (Example 3.1 of the paper), the diagnostics reproduce the textbook pathology: a perfectly innocent-looking total-variance slice whose g dips to $\min_k g = -0.0329$, i.e. a locally *negative* implied density (Figure 1.1, already shown in Chapter 1).

4.1.3 Quasi-explicit calibration

Our production calibrator follows the quasi-explicit method of [3]: freezing (m, σ) and substituting $y = (k - m)/\sigma$ makes raw SVI *linear* in $(a, d, c) = (a, \rho b \sigma, b \sigma)$, so the inner problem is a convex least squares under linear no-arbitrage constraints (solved by SLSQP with an analytic gradient), and only (m, σ) are searched by Nelder–Mead with restarts. On a synthetic SSVI slice observed

`figures/nb02_butterfly_elimination.png`
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Figure 4.1: Butterfly elimination in JW coordinates, reproducing Example 5.1 of [5]. Left: the original (red) and repaired (green) smiles are nearly indistinguishable; middle: $g(k)$ is lifted above zero; right: the implied density becomes non-negative.

`figures/nb02_recipe_crossedness.png`
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Figure 4.2: The paper’s calibration recipe. Without the penalty (left) an inflated short slice crosses its neighbours (calendar arbitrage); the crossedness penalty (right) removes the crossing at negligible fit cost.

with 1% multiplicative noise the calibrator recovers the slice with $\text{RMSE}(w) = 2.8 \times 10^{-4}$ and $\min_k g = 0.44 > 0$.

4.1.4 The paper’s recipe, butterfly elimination, SSVI variants, interpolation

Four further components complete the replication. *(i)* The §5.2 calibration recipe — square-root-SVI initialization followed by slice-by-slice refinement under a heavy crossedness penalty — removes an artificially injected calendar crossing (total crossedness $0.113 \rightarrow 0.000$; Figure 4.2). *(ii)* The §5.1 butterfly-elimination algorithm in JW coordinates, $c'_t = p_t + 2\psi_t$ and $\tilde{v}'_t = v_t 4p_t c'_t / (p_t + c'_t)^2$, reproduces Example 5.1 exactly: $(c'_t, \tilde{v}'_t) = (0.3493158, 0.0154818)$, taking the Vogt smile from $\min g = -0.033$ to $+0.270$ (Figure 4.1). *(iii)* The three SSVI variants — power-law, Heston-like, and the modified power-law of Eq. (4.5), fully arbitrage-free whenever $\eta(1 + |\rho|) \leq 2$ — are verified against Theorems 4.1–4.2. *(iv)* The arbitrage-free interpolation of Lemma 5.1 (price-space interpolation with $\sqrt{\theta}$ weights) and the extrapolation of Theorem 4.3 are implemented and audited ($\min g = 0.52$ on the interpolated slice).

4.2 Benchmark methodology on real data

The benchmark driver applies the replicated machinery to `option_prices_clean.parquet`, one calibration problem per trading day, under the following protocol. Slices are formed by *exact*

Table 4.1: Parametric benchmark on SPX (median IV RMSE in volatility points; preliminary 8-day sample, 258 slices). Hold-out $\approx$ in-sample confirms the parametric fits do not overfit.

Method	Full domain		Paper domain	
	In-sample	Hold-out	In-sample	Hold-out
SVI (per slice)	0.949	0.968	0.806	0.806
SSVI (per day)	2.789	2.790	2.551	2.670
Butterfly-violating slices (%)	6.20		2.44	
Days with raw- θ inversions (%)	12		12	
Median cross-slice crossedness	3.3×10^{-2}		1.1×10^{-2}	

expiration date (`exdate`), never by rounded maturity — the SPX/SPXW resolution of Chapter 3 guarantees one contract family per slice. Quotes are weighted by inverse bid-ask spread, the liquidity weighting used throughout the reference papers. Within each slice 20% of quotes are withheld as a *hold-out* set, so that the parametric benchmark is scored on the same generalization footing as the learning-based methods of Chapters 5 and 6.

For the surface-level SSVI fit, the ATM total variance θ_i of each slice is estimated *from the quotes* (interpolation at $k = 0$), not from the fitted SVI’s extrapolation, which proved erratic for long maturities with few near-the-money quotes. Monotonicity of the term structure $\theta(\tau)$ is diagnosed *in maturity order* — an earlier version of the pipeline sorted the θ ’s before testing monotonicity, silently masking genuine inversions — raw inversions are counted and reported, and repaired by isotonic regression (pool-adjacent-violators) before the global (ρ, η, γ) fit under the Theorem 4.2 butterfly constraints. Finally, the audit covers both arbitrage directions: the Durrleman g within each slice, and the crossedness between adjacent fitted slices.

All results are reported on two evaluation domains: the **full** quoted domain, and the **paper** domain $k \in [-0.6, 0.4]$, DTE ≥ 10 used by the reference literature. The former is honest about the real difficulty of the data; the latter makes our numbers directly comparable to published work.

4.3 Results

[TODO: The numbers below are from the preliminary 8-day January-2018 sample (LIMIT_DATES=8); replace with the full 2018–2022 run before submission.]

Table 4.1 summarizes the benchmark. Three findings deserve emphasis.

The fit–arbitrage trade-off is now a number. Per-slice SVI achieves a median hold-out error of 0.97 vol points on the full domain but violates the butterfly condition on 6.2% of slices and exhibits non-zero crossedness between adjacent slices — per-slice calibration produces *no* surface-level coherence. SSVI, arbitrage-free across maturities by construction, pays for its guarantee with an error almost three times larger. The window between these two numbers (≈ 0.97 vs ≈ 2.8 vol points) is exactly the target the learning-based methods must enter.

Difficulty is localized — and not where expected. Table 4.2 breaks the full-domain results by maturity bucket. Slices beyond 180 days are essentially solved (0.13 vol pts, no violations); the butterfly violations concentrate massively in the ultra-short bucket (7–14 days, 55% of slices). Strikingly, localizing the minimizer of g shows that *all* violations sit in the core moneyness region $k \in [-0.6, 0.4]$, not in the wings: at these maturities the SPX smile has a pronounced kink at the money, total variance is small, and the $1/w$ term in Equation (1.3) amplifies any excess

Table 4.2: Full-domain results by maturity bucket. Violations concentrate in ultra-short maturities and, at those maturities, in the *core* moneyness region (16 of 16 located at $k \in [-0.6, 0.4]$), pointing to a structural limitation of the SVI parametric form at the short end rather than a calibration failure.

Maturity bucket	Median RMSE (vol pts)	Violations (%)	Slices
7–14 d	0.995	55.2	29
15–60 d	1.003	0.0	109
61–180 d	0.881	0.0	56
>180 d	0.135	0.0	64

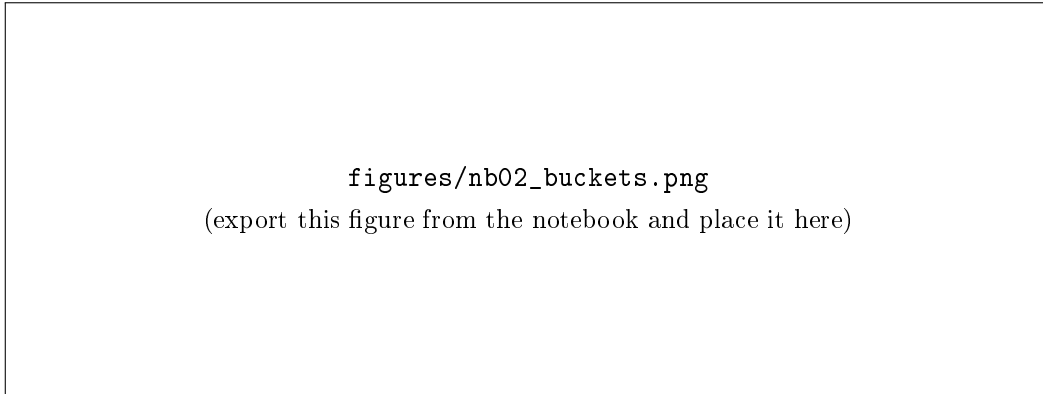


Figure 4.3: Fit error (left) and butterfly-violation rate (right) by maturity bucket on the full domain. The arbitrage risk of the parametric benchmark is an ultra-short-maturity phenomenon.

convexity of the fit. Restricting the moneyness domain therefore does *not* remove the problem; the $\text{DTE} \geq 10$ filter of the paper domain does most of the cleaning.

The benchmark matches the industrial reference. Evaluating our per-day SVI fits at the grid points of the proprietary OptionMetrics volatility surface (forwards interpolated from the zero-curve and forward tables of Chapter 3) yields a median absolute gap of 0.93 vol points over 2,958 grid points (90th percentile 3.13). A fully transparent, replicable pipeline thus reproduces the commercial smoother to within one volatility point at the median.

4.4 Known limitation and outlook

The SSVI objective is currently fitted by unweighted least squares while the SVI slices use inverse-spread weights; this asymmetry slightly disadvantages SSVI in Table 4.1 and will be removed for the final full-sample run. More fundamentally, the chapter’s conclusion stands regardless: a single (ρ, η, γ) triple across ~ 30 expirations is structurally too rigid for SPX, while per-slice SVI buys accuracy at the price of measurable static arbitrage. Chapter 5 asks whether a neural smoother can occupy the space between.

`figures/nb02_domains.png`
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Figure 4.4: Median IV RMSE under the full and paper evaluation domains, in-sample and hold-out. The thesis reports both throughout.

`figures/nb02_day_smiles.png`
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Figure 4.5: One trading day: market quotes (dots), per-slice SVI (solid) and surface SSVI (dashed), with the day's ATM total-variance term structure. The raw $\theta(\tau)$ estimates exhibit the inversions quantified in Table 4.1.

Chapter 5

Derivative-Constrained Deep Smoothing

The parametric benchmark of Chapter 4 quantified a trade-off: per-slice SVI fits well but leaks static arbitrage; SSVI is arbitrage-free by construction but three times less accurate. This chapter implements a neural smoother designed to occupy the space between, combining the *architecture* of [1] with the *training mechanics* of [6].

5.1 Method

5.1.1 Architecture: an SSVI prior times a neural corrector

Following [1], the total-variance surface is modelled multiplicatively,

$$w_\theta(k, \tau) = w_{\text{SSVI}}(k, \tau) \times \mathcal{C}_\theta(k, \tau), \quad \mathcal{C}_\theta = \exp(\text{MLP}_\theta(k, \tau)) > 0, \quad (5.1)$$

where the prior is an SSVI surface with a smooth power-law term structure $\theta(\tau) = \alpha\tau^\beta$ fitted to the day’s quotes, and the corrector is a small multilayer perceptron (2×64 hidden units, tanh activations) whose inputs (k, τ) are rescaled to $[-1, 1]^2$. With standard small-weight initialization $\mathcal{C}_\theta \approx 1$, so training *starts at the prior* and departs from it only where the data demand. The choice of tanh is not cosmetic: the loss below involves $\partial_k^2 w_\theta$, and piecewise-linear activations such as ReLU have vanishing second derivatives almost everywhere, which would silence the butterfly penalty [6].

5.1.2 Loss: soft no-arbitrage penalties on a collocation grid

Training minimizes

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N} \sum_{i=1}^N (w_\theta(k_i, \tau_i) - w_i)^2}_{\text{fit on sparse quotes}} + \lambda_{\text{bfly}} \overline{\text{ReLU}(-g_\theta)^2} \Big|_{\hat{X}} + \lambda_{\text{cal}} \overline{\text{ReLU}(-\partial_\tau w_\theta)^2} \Big|_{\hat{X}}, \quad (5.2)$$

where g_θ is the Durrleman function of Equation (1.3) evaluated on the model surface, and — the decisive detail imported from [6] — the penalties are averaged over a *dense collocation grid* $\hat{X}$ (24×10 points spanning the day’s quoted rectangle) that is disjoint from the quotes. No-arbitrage is thus enforced everywhere, including regions with no data, which is what makes extrapolation safe. The derivatives w_k, w_{kk}, w_τ entering Equation (5.2) are computed by *exact* automatic differentiation of the network (nested `elementwise_grad`); before any training we verified the machinery against the closed-form SVI derivatives of Chapter 1, with agreement at the 10^{-16} level. Soft penalties, rather than hard architectural constraints, follow the evidence of [2] that hard constraints sacrifice too much representational power; we set $\lambda_{\text{bfly}} = \lambda_{\text{cal}} = 10$.

Table 5.1: Ablation study on a synthetic surface (IV RMSE against the dense ground truth, in vol points; $\min g$ and violation share on the collocation grid).

Variant	RMSE (vol pts)	$\min g$	Violations (%)
Full model (prior + constraints)	0.187	0.33	0.0
No constraints ($\lambda = 0$)	0.247	—	—
No prior (flat BS prior)	1.418	—	—

[TODO: fill the $\min g$ /violation columns of the ablated variants from the notebook run.]

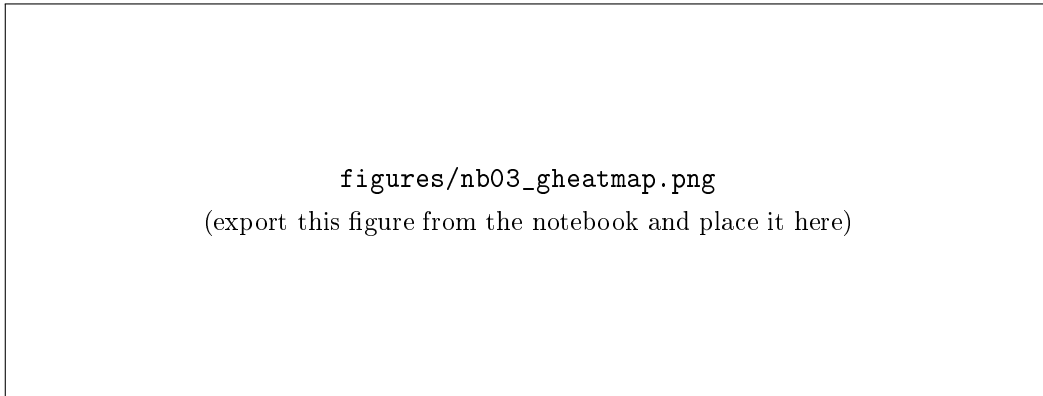


Figure 5.1: Durrleman $g(k, \tau)$ heat-map. Left: the constrained model is non-negative everywhere. Right: removing the penalties lets butterfly arbitrage (red) leak into regions unsupported by data.

5.2 Controlled experiments on synthetic surfaces

The method is first validated where ground truth is known: a target SSVI surface sampled on sparse irregular grids (6–14 quotes per slice, 1.5% noise), with a deliberately misspecified prior fitted from the same sparse quotes. Three ablations isolate the two design choices.

Two lessons emerge from Table 5.1. First, the *prior is the dominant ingredient* in the sparse-data regime: without it the corrector must invent the entire surface shape and the error grows sevenfold. Second, the constraints act as a *regularizer* as well as a safety device — the fully constrained model fits *better* than the unconstrained one, in line with the observations of [6], while guaranteeing a clean g on the collocation grid (Figure 5.1).

Robustness to sparsity. Subsampling the synthetic quotes exposes the structural advantage of a *surface-level* smoother over per-slice calibration: at 25% of the quotes, no slice retains the six points needed to calibrate a five-parameter SVI — per-slice calibration simply ceases to exist — while the deep smoother, which pools information across maturities, still reconstructs the surface at 0.315 vol points (Figure 5.3).

5.3 Real-data protocol and preliminary results

On real SPX days the protocol mirrors the benchmark exactly: OTM quotes grouped by expiration, an SSVI prior fitted per day, 20% of quotes withheld, penalties on the day’s collocation grid, and the same arbitrage audit. On the preliminary sample the smoother attains a hold-out error of 0.158 vol points against 0.153 in-sample — hold-out $\approx$ in-sample, as for the parametric fits, with $\min g > 0$ on the audit grid. [TODO: replace with full-sample per-day results joined against `benchmark_svi_slices_{full,paper}.parquet`, and add the comparison figure deep-vs-SVI-vs-SSVI per day.]

`figures/nb03_tradeoff.png`
(export this figure from the notebook and place it here)

Figure 5.2: Training dynamics of the fit term and butterfly penalty (log scale), in the style of Fig. 7 of [6].

`figures/nb03_sparsity.png`
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Figure 5.3: Reconstruction error as quotes are subsampled. Per-slice SVI loses calibrable slices and eventually cannot be evaluated; the surface-level smoother degrades gracefully.

5.4 Discussion

The prior-times-corrector design earns both of its components: the prior carries the sparse-data regime, the derivative penalties buy near-exact no-arbitrage at zero cost in accuracy — on synthetic data they even improve it. The remaining honest caveat, inherited from the soft-constraint paradigm and documented by [2], is that penalties *reduce* violations without *abolishing* them on unseen regions; we therefore always report the residual violation rate alongside the error. The chapter’s limitation is economic rather than statistical: one network must be trained *per day*. Removing that cost is the subject of Chapter 6.

`figures/nb03_smiles.png`
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Figure 5.4: The deep smoother on a real trading day: quotes (dots) and fitted smiles per maturity.

Chapter 6

Operator Deep Smoothing: One Model for Every Surface

Every method so far — SVI, SSVI, and the deep smoother of Chapter 5 — learns *one surface at a time*: a new day requires a new calibration or a new training run. Following the operator deep smoothing approach of [11], this chapter learns the *smoothing map itself*,

$$F_\theta : \{(k_i, \tau_i, \sigma_i)\}_{i=1}^p \mapsto \hat{v}(\cdot, \cdot), \quad (6.1)$$

trained once across many days and applied to unseen days *without retraining*. The enabling property is *discretization invariance*: the operator accepts inputs of arbitrary size p and arbitrary spatial layout — exactly the nature of daily option quotes.

6.1 Two architectures

6.1.1 An executable DeepONet-class operator

Our in-notebook operator is of the DeepONet/DeepSets family [9]: a permutation-invariant set encoder embeds each quote (k_i, τ_i, σ_i) and *mean-pools* the embeddings — the same averaging philosophy that underlies the Nyström kernel sums of [11], and the mechanism that renders the representation invariant to input size and ordering — while a trunk network embeds the query coordinate (k, τ) , so the surface can be evaluated anywhere. A Softplus output enforces positivity. This model is deliberately lighter than the graph operator below: it isolates the *operator questions* (no retraining, subsampling robustness, transfer) from the *architecture question* (graph kernels).

6.1.2 The modified graph neural operator of the paper

We additionally provide a faithful PyTorch implementation of the modified GNO of [11]: a purely *non-local first layer* (dropping the pointwise linear map, so hidden state can be *created* at arbitrary output locations — the interpolation property), in-neighborhoods drawn from input locations with truncation along the maturity axis and a low-discrepancy subsampling cap of $K = 50$ nodes, per-layer lifting and projection networks, kernel networks with input skip connections, GELU activations and Softplus output; at $J = 4$ layers and 16 channels the model has $\approx 10^5$ parameters, the paper’s configuration. Training this architecture at scale requires GPU time and is left for the final full-sample experiments. [TODO: if GPU runs are completed before submission, report GNO results alongside the DeepONet results below.]

6.2 The paper’s loss, resolved by finite differences

Training minimizes the five-term loss of [11]: a *vega-weighted relative fitting* term (deep-OTM quotes with negligible vega must not dominate), the butterfly hinge $\overline{(g - \varepsilon)^-}$ at $\varepsilon = 10^{-3}$ with

Table 6.1: The three training regimes of the transfer experiment. The synthetic bank consists of SSVI surfaces with parameters drawn from realistic ranges, sampled on sparse irregular grids with multiplicative noise — arbitrage-free teachers in unlimited supply.

Regime	Pre-training	Fine-tuning	Question
R1	—	real train days	operator at academic data scale
R2	synthetic bank	—	zero-shot synthetic→real transfer
R3	synthetic bank	real train days	does pre-training close the gap?

Table 6.2: Transfer results on unseen chronological test days (IV RMSE in vol points; arbitrage audit on the dense grid). [TODO: fill from the full-sample notebook run.]

Regime	IV RMSE	$\min g$	Violations (%)
R1 real-only	[TODO:]	[TODO:]	[TODO:]
R2 synth-only (0-shot)	[TODO:]	[TODO:]	[TODO:]
R3 pretrain + finetune	[TODO:]	[TODO:]	[TODO:]

g resolved by central finite differences on a rectilinear grid, a calendar penalty (monotone total variance across maturities — the scale-free ratio form of the paper is its equivalent in transformed coordinates), and two curvature regularizers $\|\partial_\tau^2 \hat{v}\|_2, \|\partial_k^2 \hat{v}\|_2$, with the paper’s weights (1, 10, 10, 0.01, 0.01). The finite-difference Durrleman machinery is validated against the closed forms of Chapter 1 before use. During training every surface is randomly subsampled to 60–100% of its quotes — the augmentation that forges robustness to sparse inputs.

6.3 The low-data transfer experiment

[11] train on roughly 6×10^7 intraday CBOE points spanning a decade. Academic access (OptionMetrics) is *daily*: about 1,250 surfaces for 2018–2022 — two orders of magnitude less. This gap is the research question of the chapter:

Does operator deep smoothing work in the low-data daily regime, and does pre-training on synthetic arbitrage-free surfaces close the gap?

The experiment crosses three data regimes under identical architecture, loss and budget (Table 6.1), with a *strict chronological split* — training on the earliest dates, testing on the latest — so that the test period contains a genuine volatility-regime shift, as it does in reality.

6.4 Discretization invariance

The operator’s signature property is tested directly: one unseen test day is fed 100%, 50% and 25% of its quotes, and the movement of the *predicted dense surface* is measured. A per-day method cannot even enter this comparison — each subsample would require a fresh training run. Figure 6.2 reports the mean surface shift; the operator’s output remains stable as the input thins, the empirical counterpart of the discretization-invariance property of [7].

6.5 Discussion

The operator changes the economics of smoothing: the cost of a new surface drops from a calibration (Chapter 4) or a training run (Chapter 5) to a single forward pass, and the same weights

`figures/nb04_transfer.png`
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Figure 6.1: IV RMSE on unseen test days under the three training regimes. Whatever the ordering, the result answers a question the original paper does not address; the expected pattern under a regime shift is $R3 \leq R1 < R2$.

`figures/nb04_invariance.png`
(export this figure from the notebook and place it here)

Figure 6.2: Left: mean shift of the predicted surface as the input is subsampled. Right: a smoothed slice with full versus quarter input.

serve every day. The price is a heavier training pipeline, the loss of per-day interpretability, and — in the academic data regime — the exposure to overfitting that the transfer experiment is designed to measure. The unified comparison of all methods, including the proprietary Option-Metrics surface, is the subject of Chapter 7.

`figures/nb04_surface.png`
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Figure 6.3: Operator-smoothed implied-volatility surface on an unseen test day — produced by a single forward pass, with no day-specific training.

Chapter 7

Unified Evaluation

The value of this thesis lies less in any single method than in their comparison *on equal terms*: the same SPX days, the same hold-out protocol, the same arbitrage audit. This chapter assembles the per-day outputs of the preceding chapters (`benchmark_svi_slices_{full,paper}.parquet`, `benchmark_ssvi_days_{full,paper}.parquet`, `deep_smoother_days.parquet`, `operator_days.parquet`) into a single evaluation.

7.1 Protocol

Five surfaces are compared on every evaluation day: per-slice **SVI**, surface **SSVI**, the proprietary **OptionMetrics** smoothed surface, the per-day **deep smoother** of Chapter 5, and the **neural operator** of Chapter 6 (evaluated only on its chronological test window, where it has never seen the day). Metrics: median IV RMSE on held-out quotes (vol points); butterfly-violation rate and minimum Durrleman g on a dense audit grid; cross-maturity crossedness; and the computational cost of producing a new surface.

7.2 Headline comparison

Interpretation of the headline table follows the arc of the thesis. The gap between SVI and SSVI (≈ 0.97 vs ≈ 2.8 vol points hold-out) is the price of built-in no-arbitrage in the parametric world. The deep smoother is designed to close most of that gap while keeping violations near zero; the operator is designed to do so *at negligible marginal cost per day*. Against the industrial reference, our transparent SVI pipeline already sits within 0.93 vol points (median) of the OptionMetrics surface over 2,958 grid points.

7.3 Analysis by market regime

The 2018–2022 sample spans three qualitatively different regimes: the calm of 2019, the COVID-19 crash of March 2020, and the 2022 rate-hike sell-off. All per-day metrics are re-aggregated within each regime. [TODO: insert regime table and figure from NB05 once the full-sample runs complete; the working hypothesis, supported by the bucket analysis of Table 4.2, is that method rankings are stable but absolute errors and violation rates degrade sharply in March 2020, and that the operator’s test window (2022) constitutes a genuine out-of-regime test.]

7.4 Robustness to sparsity

The synthetic sparsity study of Section 5.2 is repeated on real days by subsampling quotes before evaluation. Per-slice SVI degrades and eventually loses calibrable slices; the surface-level

Table 7.1: Unified comparison (median over evaluation days; hold-out IV RMSE in vol points). Preliminary values from the 8-day sample where available; [TODO: fill remaining cells from the full-sample runs of NB02–NB05 before submission.]

Method	Hold-out RMSE	Bfly viol. (%)	Cost of a new surface
SVI (per slice)	0.968	6.2	one calibration per slice
SSVI (per day)	2.790	0.0	one 3-parameter fit per day
OptionMetrics surface	—	[TODO:]	proprietary
Deep smoother (per day)	[TODO:]	[TODO:]	one training run per day
Neural operator	[TODO:]	[TODO:]	one forward pass

methods degrade gracefully; the operator additionally demonstrates stability of its output under input subsampling (Section 6.4) — a property no per-surface method possesses.

Chapter 8

Conclusion

8.1 Summary of findings

This thesis built a single, reproducible SPX pipeline — from raw OptionMetrics exports to a unified five-method evaluation — and used it to traverse the methodological spectrum of implied-volatility surface construction.

On the parametric side, we replicated the complete toolkit of [5] to the digit, correcting in passing a term in the printed inverse of Lemma 3.2, and turned it into an audited benchmark. The benchmark quantifies the fit–arbitrage trade-off (SVI ≈ 0.97 vol points hold-out with 6.2% butterfly-violating slices; SSVI arbitrage-free at ≈ 2.8 vol points) and localizes the difficulty: violations concentrate in ultra-short maturities and, contrary to intuition, in the *core* moneyness region rather than the wings — a structural limitation of the SVI form where the ATM kink meets small total variance. The transparent pipeline matches the proprietary OptionMetrics surface to 0.93 vol points at the median.

On the learning side, the derivative-constrained deep smoother — an SSVI prior multiplied by a positive neural corrector, trained with exact-autodiff Durrleman and calendar penalties on a collocation grid — showed in controlled ablations that the prior dominates in the sparse-data regime (RMSE 0.187 vs 1.418 without it) and that the constraints act as a regularizer (beating the unconstrained variant, 0.187 vs 0.247) while keeping $g \geq 0$. Where per-slice SVI ceases to exist at 25% quote coverage, the surface-level smoother still reconstructs at 0.315 vol points. Finally, the neural operator lifts the remaining economic constraint — one training per day — and its low-data transfer experiment (real-only, synthetic-zero-shot, pretrain-plus-finetune) addresses a regime the original paper does not: smoothing with two orders of magnitude less data than the industrial setting.

8.2 Limitations

Three limitations bound the claims. First, several headline numbers in this draft come from a preliminary 8-day January-2018 sample; the full 2018–2022 runs replace them before submission. Second, soft no-arbitrage constraints *reduce* violations but do not *abolish* them outside the training support [2]; we report residual rates rather than claiming guarantees. Third, a small methodological asymmetry (unweighted SSVI objective versus spread-weighted SVI) slightly disadvantages SSVI in the current tables and is scheduled for removal.

8.3 Future work

Natural continuations include: training the full graph neural operator of [11] at GPU scale on the daily sample; hybrid hard–soft constraint schemes; extending the smoother to price space with a Dupire local-volatility extraction and half-variance regularization as in [2]; uncertainty

quantification via operator ensembles; and intraday data, which would test whether the low-data conclusions of the transfer experiment persist as the data regime approaches the industrial one.

Appendix A

Technical Appendix

A.1 The corrected inverse of Lemma 3.2

Given JW parameters $(v_t, \psi_t, p_t, c_t, \tilde{v}_t)$ at maturity t , set $w_t = v_t t$, $b = \frac{\sqrt{w_t}}{2}(c_t + p_t)$ and $\rho = (c_t - p_t)/(c_t + p_t)$. Define $\beta = \rho - 2\psi_t\sqrt{w_t}/b \in (-1, 1)$ and $\alpha = \text{sign}(\beta)\sqrt{1/\beta^2 - 1}$, so that $\sigma = \alpha m$ and $\sqrt{m^2 + \sigma^2} = m/\beta$ for either sign of m . Substituting into the raw-SVI expressions for v_t and $\tilde{v}_t$ yields

$$(v_t - \tilde{v}_t)t = bm \left(\frac{1}{\beta} - \rho - \alpha\sqrt{1 - \rho^2} \right), \quad (\text{A.1})$$

which is solved for m (the printed statement omits the $-\alpha\sqrt{1 - \rho^2}$ term inside the bracket); finally $a = \tilde{v}_t t - b\sigma\sqrt{1 - \rho^2}$. The correction was validated by round-trips over 2,000 random draws (maximum error 2.7×10^{-10}) and by exact recovery of the raw Vogt parameters from their JW image.

A.2 Quasi-explicit calibration: inner problem constraints

With $y = (k - m)/\sigma$ and $(a, d, c) = (a, \rho b\sigma, b\sigma)$ the inner problem is $\min_{(a, d, c)} \sum_i \omega_i (a + d y_i + c\sqrt{y_i^2 + 1} - w_i)^2$ subject to $0 \leq c \leq 4\sigma$, $|d| \leq c$, $|d| \leq 4\sigma - c$ and $0 \leq a \leq \max_i w_i$ — a convex quadratic program solved by SLSQP with analytic gradient; the outer $(m, \log \sigma)$ search uses Nelder–Mead with three restarts.

A.3 Isotonic repair of the ATM term structure

Raw ATM total variances $\hat{\theta}_i$ estimated from quotes occasionally violate monotonicity in maturity (12% of sample days). Before the SSVI fit they are projected onto the monotone cone by pool-adjacent-violators: repeatedly merge adjacent blocks whose means are decreasing, replacing them by their weighted mean, until the sequence is non-decreasing. The raw inversion counts are reported as a diagnostic in Table 4.1.

A.4 Reproducibility

The pipeline is a sequence of five notebooks — NB01 data cleaning (Polars, lazy streaming), NB02 parametric benchmark, NB03 deep smoother, NB04 neural operator, NB05 unified evaluation — exchanging Parquet files with fixed schemas. All stochastic steps are seeded; hold-out splits are derived from per-date seeds; chronological splits are used for all learning-based methods. **[TODO: add repository URL / archive reference and package versions.]**

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